

Two-Lagrangian CL inflow fits of the SPARC category-B galaxies: detection, model selection, fit quality and robustness

Automated analysis for E.P.J. de Haas

Summary

Category B contains the 45 SPARC galaxies whose single-Lagrangian (single-L) CL residuals show a significant $+-+$ wave. For all of them the two-Lagrangian (two-L) model of Eqs. (24)–(26) was fitted and compared with the single-L fit by BIC. In 23 galaxies the two-L description is accepted ($\Delta\text{BIC} < -6$); in 22 the wave is real in sign but too weak to justify two extra parameters, and the single-L fit is kept. The accepted fits split into three groups. 13 are clean nested fits: physically ordered ($M_2 > M_1$, no large drop at R_2) and statistically adequate. 5 are improved but still need more regions ($\chi_\nu^2 > 2$). 5 are not nested, meaning the fit exploits the discontinuity at R_2 . For the clean nested fits the median RMS_{rel} in v^2 falls from 0.251 to 0.127 and the median Durbin–Watson statistic rises from 0.33 to 1.33. The outer scale is well determined ($\sigma_{v_{L2}}/v_{L2} = 1.6\%$, $\sigma_{R_2}/R_2 = 5\%$ from profile likelihood). The inner scale is less well determined ($\sigma_{M_1}/M_1 = 22\%$). Eight of the 13 galaxies identified as double fits in 2018 are recovered as clean nested fits. The main caveats are that R_2 is a discontinuity location, whose likelihood is non-smooth and in some galaxies multimodal, and that residual structure persists in 7 of the 13 clean fits, pointing to further (virial) regions.

1 Sample and procedure

Selection. Category B was defined in the full-SPARC pipeline. After the single-L CL fit ($H_z = 2.2 \times 10^{-18} \text{ s}^{-1}$, fits on v^2 with $\sigma_{v^2} = 2V_{\text{obs}}\delta V_{\text{obs}}$), the weighted residuals must fail the one-sided runs test ($p_{\text{runs}} < 0.05$), agree to at least 80% with a three-segment $+-+$ sign template, and agree better with it than with the reverse $-+-$ template.

Two-L model. Eqs. (24)–(26): bulge $r \leq R_1$ and bar zone $R_1 < r \leq R_2$ governed by L_1 (M_1, R_1), disk $r > R_2$ by L_2 (M_2, R_2), with asymptotic speeds $v_{Li} = \sqrt{3/2} X_i$, $X_i = \sqrt{2GM_i/R_i} - H_z R_i$. A shared offset Φ_{BH} is tested as a third variant.

Fitting and selection. The two-L fit is seeded at $R_2 = r_{\times 1}/0.8$ (first $+\rightarrow-$ residual crossover) and cross-checked by a global grid and by the R_2 profile scan below; the best optimum is kept. A two-L fit is accepted if $\Delta\text{BIC} = \text{BIC}_{2\text{L}} - \text{BIC}_{1\text{L}} < -6$ (strong evidence), and marked tentative when $n \leq 13$.

Uncertainties. M_1, R_1, M_2 : local covariance. R_2 : because v^2 is discontinuous at R_2 , $\chi^2(R_2)$ is piecewise smooth, changing slope whenever R_2 crosses a data radius, so the covariance error is unreliable. R_2 and M_2 are therefore given as profile-likelihood intervals ($\Delta\chi^2 \leq 1$, other parameters refitted on a grid of 140 R_2 values). Jackknife: leave-one-out refits. Systematic: refits with distance $+\delta D$ and inclination $+\delta i$, added in quadrature. H_z sensitivity: refit with $H_z = 0$.

Post-hoc classification of accepted fits. **Not nested (N):** $M_2 < M_1$ or v^2 drops by more than 40% at R_2 . **Needs more regions (M):** $\chi_\nu^2 > 2$ after two-L. **Clean nested:** neither. Grades: **I** all four parameters within 30% (profile errors for R_2, M_2), at least 2 points in bulge and bar zones and 3 in the disk, jackknife scatter $\leq 30\%$; **II** both v_{L1} and v_{L2} within 10%; **III** otherwise.

2 Model selection outcome

All 45 galaxies have significant single-L residual structure ($p_{\text{runs}} < 0.05$ by selection). The rejected galaxies differ from the accepted ones mainly in amplitude, not in shape. Their single-L fits are already good (median $\chi_\nu^2 = 0.50$, $\text{RMS}_{\text{rel}} = 0.133$, against 2.98 and 0.247 for the accepted ones), so the wave is coherent but lies

Table 1: Category-B outcome.

Outcome	N	Galaxies
Clean nested two-L, grade I	6	D631–7, DDO161, IC2574, NGC3741, NGC7793, UGC04278
Clean nested two-L, grade II	7	ESO116–G012, F571–8, NGC0055, NGC0247, UGC00731, UGC07524, UGC12732
Two-L, needs more regions (M)	5	DDO154, NGC1003, NGC2403, NGC5585, UGC03580
Not nested (N)	5	NGC3726, NGC4013, UGC02885, UGC06614, UGC06787
Single-L kept, $-6 \leq \Delta\text{BIC} < -2$	7	NGC3109, NGC3972, NGC4010, UGC06446, UGC07125, UGC07323, UGC07399
Single-L kept, $-2 \leq \Delta\text{BIC} < 0$	3	ESO079–G014, NGC4214, UGC05829
Single-L kept, $\Delta\text{BIC} \geq 0$	12	F568–1, F568–3, F574–1, F579–V1, KK98–251, NGC0100, NGC2976, NGC5005, UGC08286, UGC08550, UGC12632, UGCA444

within the SPARC errors. 12 of them have $\Delta\text{BIC} \geq 0$. Among these are F568-1, F574-1, F579-V1 and UGC 8286, which were single fits in 2018 as well. The seven galaxies with $-6 \leq \Delta\text{BIC} < -2$ (positive but not strong evidence) include the 2018 double fits NGC 3109, NGC 3972 and UGC 6446. With $n = 10$ –25 points, a threshold of -6 is conservative for them, and they are the natural candidates for confirmation with more data. Their status is therefore *undecided* rather than single-L.

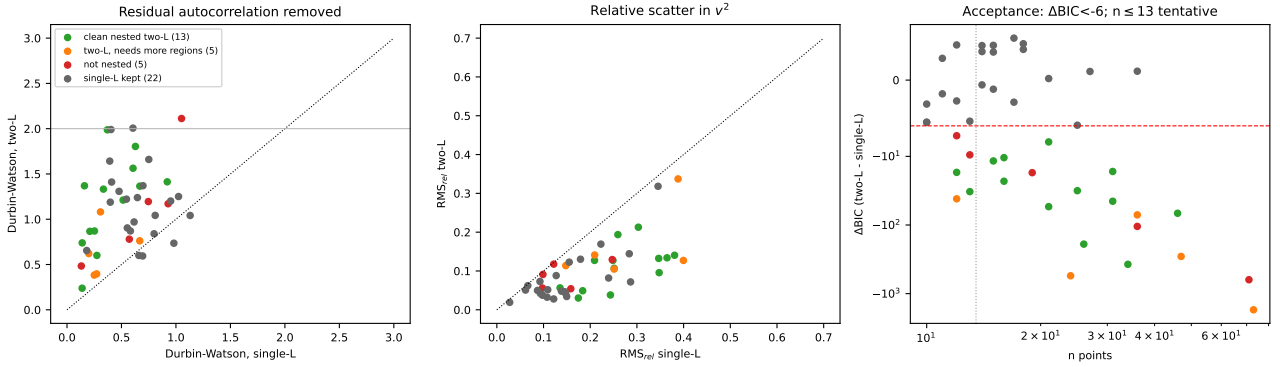


Figure 1: Left: Durbin–Watson statistic before and after the two-L fit (2 = uncorrelated residuals). Middle: relative RMS in v^2 , single-L versus two-L. Right: ΔBIC versus number of points; acceptance threshold -6 (red) and tentative limit $n \leq 13$ (dotted). Colours: clean nested (green), needs more regions (orange), not nested (red), single-L kept (grey).

3 Fit quality of the accepted fits

Table 2: Fit-quality statistics (medians), single-L $\rightarrow$ two-L.

	Clean nested (13)	Needs more regions (5)	Not nested (5)
χ^2_ν	2.57 $\rightarrow$ 0.39	9.69 $\rightarrow$ 3.48	2.98 $\rightarrow$ 2.19
RMS_{rel}	0.251 $\rightarrow$ 0.127	0.251 $\rightarrow$ 0.127	0.122 $\rightarrow$ 0.091
Durbin–Watson	0.33 $\rightarrow$ 1.33	0.27 $\rightarrow$ 0.62	0.75 $\rightarrow$ 1.17
$p_{\text{runs}} < 0.05$ after two-L	7/13	4/5	4/5
$p_{\text{runs}} < 0.05$ after two-L + Φ_{BH}	5/13	4/5	3/5
ΔBIC (median)	-31.6	-286.9	-17.3

Clean nested fits. The two-L model removes most of the coherent residual. The median RMS_{rel} halves, the median χ^2_ν drops from 2.57 to 0.39, and the residual sign pattern is broken in all cases (atlas). Residual structure is not fully removed in 7 galaxies (DDO161, NGC0055, NGC0247, NGC7793, UGC04278, UGC07524, UGC12732). Adding the shared offset Φ_{BH} brings p_{runs} above 0.05 in DDO 161 and UGC 12732. In the others the remaining pattern is consistent with a further region (virial window) as introduced in the H_z paper; in NGC 7793 it is the declining outer curve. As in category A, $\chi^2_\nu < 1$ is common (median 0.39; 7 galaxies with $P(\chi^2 \leq \chi^2_{\text{obs}}) < 0.01$) because SPARC errors are conservative. Shapiro–Wilk rejects normality of the two-L residuals in 2 of 13.

Offset Φ_{BH} . Among the clean fits only IC 2574 strongly prefers the offset ($\Delta\text{BIC} = -7.9$, $\Phi_{\text{BH}} = 23 \pm 6 \text{ km}^2 \text{ s}^{-2}$). ESO116-G012, NGC 247 and UGC 12732 show positive evidence ($-6 < \Delta\text{BIC} < -4.7$; $\Phi_{\text{BH}} \approx 430$, 360 and $950 \text{ km}^2 \text{ s}^{-2}$). NGC 247 and UGC 12732 are the two galaxies where the 2018 fit also needed V_0^2 (359 and 949), in remarkable agreement.

Needs more regions. DDO154, NGC1003, NGC2403, NGC5585, UGC03580. The two-L fit improves these enormously (NGC 2403: $\Delta\text{BIC} = -1719$), but χ_ν^2 stays at 2.8–7.9 and the atlas shows coherent outer residuals. These are three-region (bulge–bar–disk–virial) or three-Lagrangian systems.

Not nested. NGC3726, NGC4013, UGC02885, UGC06614, UGC06787. Here the optimiser places R_2 so that v^2 drops by 45–85% or $M_2 < M_1$. In NGC 4013 the inner radius also runs to its lower bound. These are mostly early types or edge-on systems (Sa–Sc; NGC 4013 is edge-on) whose rising–peaked–declining curves are not a bar-in-disk configuration. The statistical gain is real but the two-L parameters have no nested-spiral interpretation, so these galaxies should be moved to category C or D.

4 Parameter determination

For the clean nested fits the outer Lagrangian is much better determined than the inner one: median relative errors are 7% (M_2), 5% (R_2 , profile) and 1.6% (v_{L2}), against 22%, 13% and 5.4% for M_1 , R_1 and v_{L1} . As in category A, each pair is strongly correlated (median $\rho_{M_1 R_1} = 0.96$), and the well-determined quantities are the two asymptotic speeds.

The R_2 likelihood. Fig. 2 shows $\Delta\chi^2(R_2)$. In most galaxies the minimum is sharp and lies between two data radii, so the $\Delta\chi^2 \leq 1$ interval is effectively the gap between neighbouring points and is often asymmetric. In NGC 247 and especially UGC 12732 the profile has several near-equal minima (UGC 12732: four $\Delta\chi^2 \leq 1$ islands between 9.6 and 10.1 kpc, plus shallow minima down to 5 kpc). Its R_2 is therefore set by the sampling rather than measured, and its grade-II status should be read with that in mind. This sampling dependence is intrinsic to a model with a discontinuity; a smooth transition of width $\sim$ the point spacing would remove it.

Systematics, jackknife, H_z . Distance and inclination systematics exceed the statistical errors (median sys/stat 1.2 for M_1 , 1.8 for M_2 , 2.6 for R_2) and should be quoted alongside them. Leave-one-out scatter is within 30% for grade I but larger for several grade II galaxies (ESO116-G012: one point controls M_1). Setting $H_z = 0$ changes no parameter by more than 1.9%.

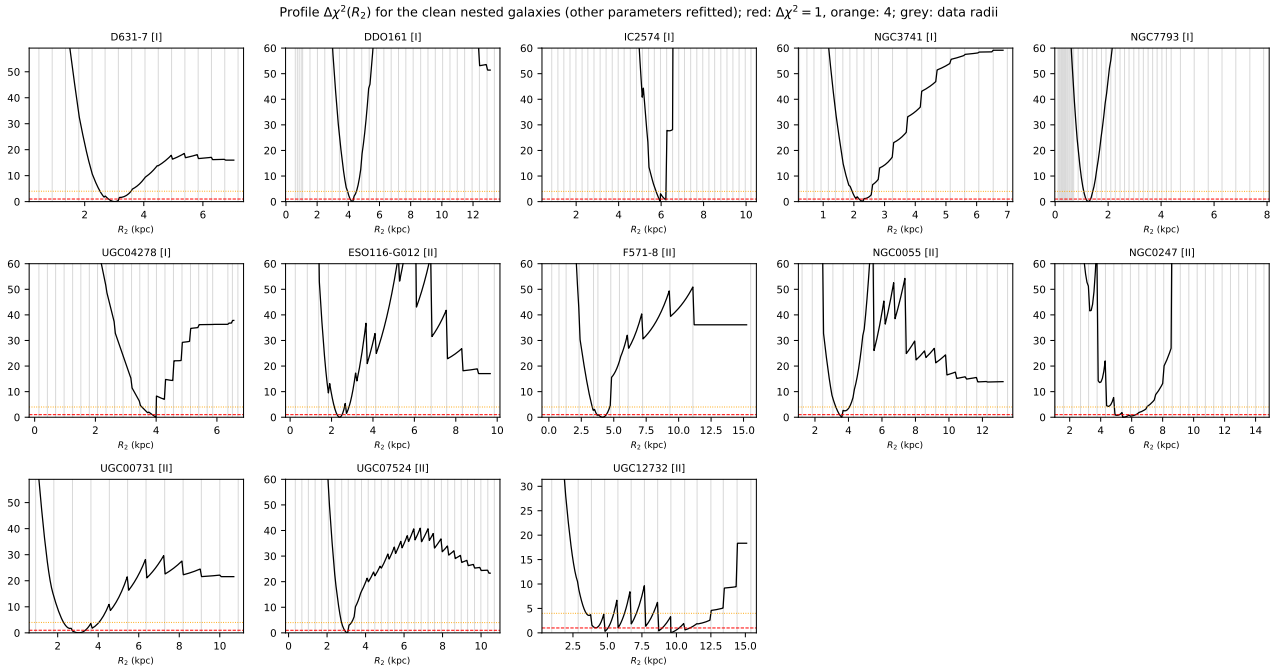


Figure 2: Profile $\Delta\chi^2(R_2)$ for the 13 clean nested fits. Vertical grey lines are the data radii; the likelihood changes slope at each, so R_2 is localised between data points. Multiple minima (NGC 247, UGC 12732) indicate sampling-limited R_2 .

5 Physical consistency

- **Seed versus fit.** $R_2/r_{\times 1} = 1.20$ (16–84%: 1.11–1.85), consistent with the seed factor $1/0.8 = 1.25$. The residual crossover is a good locator of the Lagrangian reset.
- **Nesting.** $R_1/R_2 = 0.29$, $M_2/M_1 = 8.4$ (16–84%: 6.2–12.7), $v_{L2}/v_{L1} = 1.63$. The bar zone lies at about $0.3\text{--}1 R_2$, and the outer Lagrangian carries about eight times the inner mass.
- **Disk scale.** $R_2/R_{\text{disk}} = 1.58$ (Spearman $\rho = 0.52$), so the reset lies about 1–4 disk scale lengths out, where bars and inner rings are found.
- **Asymptotic speed.** v_{L2} tracks SPARC V_{flat} ($r = 0.992$, 11 galaxies), with $v_{L2}/V_{\text{flat}} = 1.14$. As in category A, the CL curve is still rising at the last measured point.
- **Mass.** $M_2/M_{\text{bar}} = 1.06$ (16–84%: 0.61–2.16), $r(\log M_2, \log M_{\text{bar}}) = 0.74$. The outer Lagrangian mass is comparable to the SPARC baryonic mass.
- **Jump at R_2 .** For the clean fits v^2 changes by -0.12 to $+0.76$ (median $+0.19$) across R_2 . No data point lies at R_2 , so the jump size is a model feature that the data do not constrain directly.
- **Morphology.** The clean fits are all late types (Sc–Im: Sc 1, Sd 4, Sm 4, Im 4), and 10 of 13 are Q=1. The non-nested and multi-region cases include all the early types.

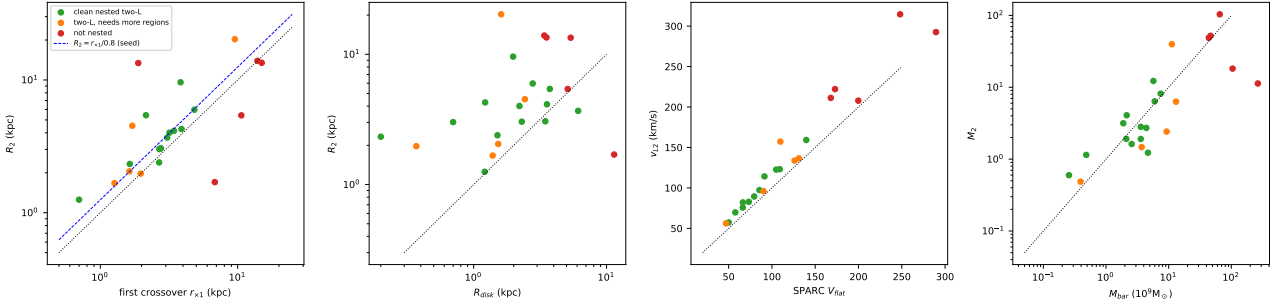


Figure 3: Accepted fits. Left to right: R_2 versus first residual crossover (dashed: seed relation); R_2 versus R_{disk} ; v_{L2} versus SPARC V_{flat} ; M_2 versus M_{bar} . Colours as in Fig. 1.

6 Assessment

Detection works. The wave screen plus BIC selects 13 clean nested two-L galaxies. All are late types with Q mostly 1. They include 8 of the 13 galaxies identified as double fits by hand in 2018, and the automated R_2 agrees with the crossover radius. In these galaxies the two-L model is preferred by ΔBIC from -8 to -374 , removes the coherent residual wave, and yields an outer Lagrangian (v_{L2} , M_2) that matches independent photometric and kinematic scales.

Grades. Grade I (D631–7, DDO161, IC2574, NGC3741, NGC7793, UGC04278) is the robust core, with every parameter determined to within 30% and each zone sampled. Grade II (ESO116–G012, F571–8, NGC0055, NGC0247, UGC00731, UGC07524, UGC12732) has well-determined asymptotic speeds, but one parameter is weakly constrained or influenced by one point. UGC 12732 and NGC 247 have sampling-limited R_2 .

Limits. (i) ΔBIC accepts any statistically better fit, so physical screening (N, M) is essential. 10 of the 23 accepted fits are not clean nested spirals. (ii) R_2 is localised only to the gap between data points, and the size of the v^2 jump is unconstrained. (iii) Residual structure remains in about half of the clean fits, mostly in the outer disk, which motivates the virial-window extension. (iv) For $n \lesssim 15$ the -6 threshold is conservative, so the seven galaxies with $-6 \leq \Delta\text{BIC} < -2$ remain undecided.

Table 3: All category-B galaxies: single-L fit, wave statistics and model-selection outcome. † = 2018 double fit.

Galaxy	Type	Q	n	$\chi^2_{\nu,1L}$	RMS _{1L}	p_{runs}	A_{+-+}	$r_{\times 1}$	$\chi^2_{\nu,2L}$	ΔBIC	Outcome	Grade
NGC2403	Scd	1	73	32.04	0.251	3.6e-13	0.96	1.27	7.93	-1719.3	two-L, needs more regions	IM
UGC06787	Sab	2	71	25.98	0.247	1.7e-13	0.87	1.89	17.26	-627.4	not nested	N
NGC5585	Sd	1	24	28.93	0.400	1.1e-04	0.92	1.63	4.05	-549.1	two-L, needs more regions	IIM
IC2574†	Sm	2	34	13.13	0.251	6.8e-06	0.94	4.84	1.30	-374.1	clean nested two-L	I
UGC03580	Sa	2	47	9.69	0.388	1.2e-10	1.00	1.71	3.29	-286.9	two-L, needs more regions	IM
NGC0247†	Sd	2	26	8.58	0.243	8.7e-06	1.00	2.15	0.43	-189.9	clean nested two-L	II
NGC4013	Sb	2	36	5.52	0.122	2.0e-07	0.92	14.98	2.36	-104.9	not nested	N
NGC1003	Scd	1	36	4.91	0.148	1.0e-04	0.92	9.54	2.77	-71.2	two-L, needs more regions	IM
NGC7793	Sd	1	46	2.57	0.303	4.4e-08	0.93	0.70	0.90	-67.5	clean nested two-L	I
NGC3741†	Im	1	21	3.52	0.365	7.5e-05	1.00	1.64	0.39	-54.0	clean nested two-L	I
DDO161†	Im	1	31	2.15	0.348	9.0e-05	0.90	3.90	0.38	-45.1	clean nested two-L	I
DDO154	Im	2	12	7.43	0.210	3.8e-02	0.83	1.97	3.48	-41.6	two-L, needs more regions (tent.)	IIM
F571-8†	Sc	1	13	3.79	0.347	4.6e-03	1.00	3.44	0.44	-32.6	clean nested two-L (tent.)	II
UGC04278†	Sd	1	25	1.97	0.381	1.4e-05	1.00	3.19	0.35	-31.6	clean nested two-L	I
UGC12732†	Sm	1	16	2.81	0.209	9.9e-04	1.00	3.84	0.90	-23.0	clean nested two-L	II
UGC02885	Sc	1	19	2.03	0.098	4.8e-03	0.89	6.82	0.75	-17.3	not nested	N
UGC00731	Im	1	12	2.34	0.175	8.3e-03	1.00	2.73	0.16	-17.1	clean nested two-L (tent.)	II
UGC07524	Sm	1	31	1.07	0.249	6.7e-05	0.94	2.76	0.28	-16.6	clean nested two-L	II
ESO116-G012	Sd	1	15	2.57	0.259	7.1e-03	1.00	2.68	1.49	-11.7	clean nested two-L	II
D631-7†	Im	1	16	1.33	0.184	2.3e-03	1.00	2.69	0.22	-10.4	clean nested two-L	I
UGC06614	Sa	1	13	1.62	0.158	4.6e-03	0.92	10.64	0.33	-9.8	not nested (tent.)	N
NGC0055	Sm	2	21	1.04	0.135	1.3e-04	1.00	3.07	0.33	-8.1	clean nested two-L	II
NGC3726	Sc	2	12	2.98	0.099	3.8e-02	0.83	13.94	2.19	-7.3	not nested (tent.)	N
NGC3109†	Sm	1	25	0.78	0.127	2.6e-04	0.96	2.06	0.26	-5.9	single-L kept	-
UGC07323	Sdm	1	10	1.68	0.284	3.6e-02	1.00	2.91	0.54	-5.6	single-L kept	-
NGC3972†	Sbc	1	10	1.42	0.138	2.2e-02	1.00	2.61	0.20	-5.5	single-L kept	-
UGC07125	Sm	1	13	1.15	0.150	7.2e-03	1.00	4.33	0.23	-5.4	single-L kept	-
UGC07399	Sdm	1	10	1.29	0.098	2.5e-02	0.90	1.84	0.43	-3.1	single-L kept	-
UGC06446†	Sd	1	17	0.67	0.147	4.3e-03	0.94	1.74	0.11	-2.9	single-L kept	-
NGC4010	Sd	2	12	1.69	0.239	3.8e-02	0.83	3.50	1.15	-2.7	single-L kept	-
UGC05829†	Im	2	11	0.83	0.286	1.3e-02	1.00	3.14	0.12	-1.8	single-L kept	-
ESO079-G014	Sbc	1	15	1.12	0.345	9.3e-03	0.93	5.65	0.72	-1.2	single-L kept	-
NGC4214	Im	2	14	0.53	0.122	9.7e-03	0.93	1.04	0.04	-0.6	single-L kept	-
NGC0100	Scd	1	21	0.37	0.146	2.5e-03	0.90	2.51	0.07	0.2	single-L kept	-
NGC2976	Sc	2	27	0.47	0.155	1.3e-02	0.85	1.01	0.28	1.1	single-L kept	-
UGCA444	Im	2	36	0.26	0.223	7.6e-06	0.97	0.97	0.09	1.2	single-L kept	-
UGC08550	Sd	1	11	0.68	0.061	1.3e-02	1.00	1.47	0.60	2.9	single-L kept	-
UGC12632	Sm	1	15	0.28	0.179	1.8e-03	1.00	3.55	0.18	3.7	single-L kept	-
F574-1	Sd	1	14	0.16	0.109	2.9e-03	1.00	2.35	0.03	3.7	single-L kept	-
F568-3	Sd	1	18	0.19	0.092	1.4e-02	0.94	5.39	0.09	4.0	single-L kept	-
F579-V1	Sc	1	14	0.10	0.108	2.7e-03	0.93	2.11	0.04	4.5	single-L kept	-
KK98-251	Im	2	15	0.18	0.066	1.3e-02	0.93	1.24	0.13	4.6	single-L kept	-
F568-1	Sc	1	12	0.05	0.094	7.7e-03	1.00	3.08	0.02	4.6	single-L kept	-
NGC5005	Sbc	1	18	0.20	0.087	1.1e-03	1.00	1.47	0.16	4.8	single-L kept	-
UGC08286	Scd	1	17	0.15	0.028	1.9e-02	0.94	1.42	0.16	5.5	single-L kept	-

Table 4: Accepted two-L fits: parameters. R in kpc, M in $10^9 M_\odot$, v_L in km s^{-1} . R_1, M_1 : statistical $\pm$ systematic; R_2, M_2 : profile $\Delta\chi^2 \leq 1$ interval, and systematic in brackets. Jump = fractional v^2 change at R_2 . Φ_{BH} from the two-L+ Φ_{BH} fit.

Galaxy	Grade	n	R_1	M_1	R_2	M_2	v_{L1}	v_{L2}	zones	jump	Φ_{BH}
IC2574	I	34	$2.55 \pm 0.054 \pm 0.13$	$0.518 \pm 0.021 \pm 0.037$	$5.97^{+0}_{-0.013}$ [0.31]	$3.16^{+0.0097}_{-0.015}$ [0.22]	50.9 ± 0.53	82.1 ± 0.48	7/11/16	+0.20	23 ± 6.2
NGC7793	I	46	$0.338 \pm 0.036 \pm 0.017$	$0.134 \pm 0.026 \pm 0.031$	$1.25^{+0.054}_{-0.053}$ [0.062]	$1.23^{+0.064}_{-0.055}$ [0.28]	71.6 ± 3.4	112.4 ± 2.4	7/14/25	+0.00	$-93 \pm 1.5e + 02$
NGC3741	I	21	$0.485 \pm 0.065 \pm 0.026$	$0.0397 \pm 0.0081 \pm 0.0027$	$2.33^{+0.13}_{-0.13}$ [0.12]	$0.599^{+0.00027}_{-0.055}$ [0.041]	32.5 ± 1.4	57.4 ± 0.89	2/7/12	+0.21	128 ± 65
DDO161	I	31	$1.26 \pm 0.16 \pm 0.38$	$0.196 \pm 0.044 \pm 0.061$	$4.27^{+0.017}_{-0.15}$ [1.3]	$1.92^{+0.012}_{-0.099}$ [0.59]	44.7 ± 2.5	75.7 ± 0.72	5/6/20	+0.18	156 ± 92
UGC04278	I	25	$1.03 \pm 0.14 \pm 0.31$	$0.238 \pm 0.053 \pm 0.071$	$4.01^{+0.094}_{-0.2}$ [1.2]	$4.08^{+0.03}_{-0.2}$ [1.2]	54.5 ± 2.7	114.2 ± 2.3	4/10/11	+0.76	16 ± 56
D631-7	I	16	$1.43 \pm 0.16 \pm 0.033$	$0.242 \pm 0.056 \pm 0.015$	$3.02^{+0.12}_{-0.2}$ [0.07]	$1.15^{+0.062}_{-0.14}$ [0.071]	46.7 ± 2.9	69.8 ± 2.4	3/3/10	+0.08	1 ± 33
NGC0247	II	26	$1.59 \pm 0.032 \pm 0.081$	$0.835 \pm 0.031 \pm 0.048$	$5.42^{+0.77}_{-0.41}$ [0.28]	$6.37^{+1.7}_{-0.62}$ [0.37]	82.2 ± 0.79	122.7 ± 4.1	1/8/17	-0.08	$359 \pm 1.3e + 02$
F571-8	II t	13	$1.14 \pm 0.15 \pm 0.23$	$0.687 \pm 0.17 \pm 0.14$	$4.13^{+0.19}_{-0.32}$ [0.83]	$8.16^{+1.3}_{-1.3}$ [1.6]	88.2 ± 5.5	159.2 ± 4.1	2/4/7	+0.32	$99 \pm 1.8e + 02$
UGC12732	II	16	$1.79 \pm 0.18 \pm 0.56$	$0.978 \pm 0.14 \pm 0.37$	$9.6^{+0.55}_{-0}$ [2.9]	$12.2^{+1.5}_{-0.014}$ [4.5]	83.8 ± 2.1	127.5 ± 2	1/9/6	-0.12	$955 \pm 2.6e + 02$
UGC00731	II t	12	$1.02 \pm 0.12 \pm 0.31$	$0.26 \pm 0.057 \pm 0.079$	$3.04^{+0.32}_{-0}$ [0.9]	$1.62^{+0.26}_{-0.01}$ [0.49]	57.2 ± 3.3	82.8 ± 2.4	1/2/9	-0.10	$411 \pm 4.5e + 02$
UGC07524	II	31	$1.06 \pm 0.14 \pm 0.054$	$0.228 \pm 0.052 \pm 0.024$	$3.06^{+0.011}_{-0.12}$ [0.15]	$1.91^{+0.011}_{-0.17}$ [0.2]	52.5 ± 2.9	89.4 ± 1.4	3/5/23	+0.25	188 ± 84
ESO116-G012	II	15	$0.927 \pm 0.13 \pm 0.28$	$0.412 \pm 0.11 \pm 0.12$	$2.39^{+0.083}_{-0.1}$ [0.84]	$2.82^{+0.14}_{-0.17}$ [0.84]	75.6 ± 5.4	123.1 ± 1.6	2/2/11	+0.19	$432 \pm 1.4e + 02$
NGC0055	II	21	$1.52 \pm 0.25 \pm 0.079$	$0.381 \pm 0.11 \pm 0.021$	$3.68^{+0.16}_{-0.16}$ [0.19]	$2.72^{+0.2}_{-0.2}$ [0.15]	56.9 ± 3.8	97.3 ± 0.77	1/3/17	+0.34	$593 \pm 2.1e + 02$
NGC5585	IIM	24	$0.234 \pm 0.015 \pm 0.07$	$0.0305 \pm 0.003 \pm 0.0093$	2.05^{+0}_{-0} [0.62]	$1.47^{+0.0076}_{-0.0021}$ [0.45]	41.0 ± 0.79	96.0 ± 0.59	1/9/14	+0.98	17 ± 52
DDO154	IIM t	12	$1.2 \pm 0.13 \pm 0.059$	$0.18 \pm 0.047 \pm 0.012$	$1.97^{+0.0054}_{-0}$ [0.098]	$0.486^{+0.005}_{-0.005}$ [0.033]	44.0 ± 3.4	56.3 ± 0.13	2/2/8	-0.09	128 ± 43
NGC2403	IM	73	$0.495 \pm 0.011 \pm 0.025$	$0.355 \pm 0.014 \pm 0.025$	1.67^{+0}_{-0} [0.085]	2.42^{+0}_{-0} [0.17]	96.2 ± 0.84	136.5 ± 0.11	4/12/57	-0.16	1028 ± 90
UGC03580	IM	47	$1.01 \pm 0.042 \pm 0.27$	$0.824 \pm 0.052 \pm 0.23$	$4.52^{+0.019}_{-0}$ [1.1]	$6.31^{+0.033}_{-0.032}$ [1.6]	102.6 ± 1.3	133.8 ± 0.44	8/21/18	-0.34	$3601 \pm 2.4e + 02$
NGC1003	IM	36	$2.12 \pm 0.12 \pm 0.69$	$1.8 \pm 0.12 \pm 0.59$	20.3^{+0}_{-0} [6.1]	$39.8^{+0.15}_{-0.15}$ [12]	104.4 ± 0.74	157.3 ± 0.97	2/22/12	-0.19	$3370 \pm 5.5e + 02$
UGC06787	N	71	$0.124 \pm 0.025 \pm 0.031$	$0.58 \pm 0.12 \pm 0.15$	13.4^{+0}_{-0} [3.3]	104^{+0}_{-0} [26]	245.4 ± 0.48	314.6 ± 0.68	1/51/19	-0.45	$24405 \pm 1.6e + 04$
NGC4013	N	36	$0.02 \pm 0.84 \pm 0.0028$	$0.0577 \pm 2.4 \pm 0.008$	13.5^{+0}_{-0} [1.9]	52^{+0}_{-0} [7.2]	192.9 ± 6	222.1 ± 1.4	0/12/24	-0.56	$19890 \pm n/a$
UGC02885	N	19	$0.863 \pm n/a \pm 0.039$	$0.42 \pm n/a \pm 0.6$	1.7^{+0}_{-0} [0.17]	$11.3^{+0}_{-0.006}$ [1.3]	$375.2 \pm n/a$	$292.6 \pm n/a$	0/1/18	-0.69	$73141 \pm 3.4e + 03$
UGC06614	N t	13	$4.28 \pm 2e + 07 \pm 0.56$	$65.8 \pm 9.1e + 08 \pm 22$	$5.41^{+0.77}_{-0.92}$ [0.54]	$18.2^{+3.6}_{-0.32}$ [6.3]	$445.0 \pm 2.1e + 09$	207.8 ± 7.1	1/0/12	-0.85	$29531 \pm 1.3e + 05$
NGC3726	N t	12	$3.57 \pm 0.27 \pm 0.5$	$8.99 \pm 1 \pm 1.3$	$13.9^{+0.92}_{-0}$ [1.9]	$48.8^{+0.47}_{-0.32}$ [7.2]	180.0 ± 4.1	211.3 ± 4.2	1/6/5	-0.45	$6415 \pm 3.6e + 03$

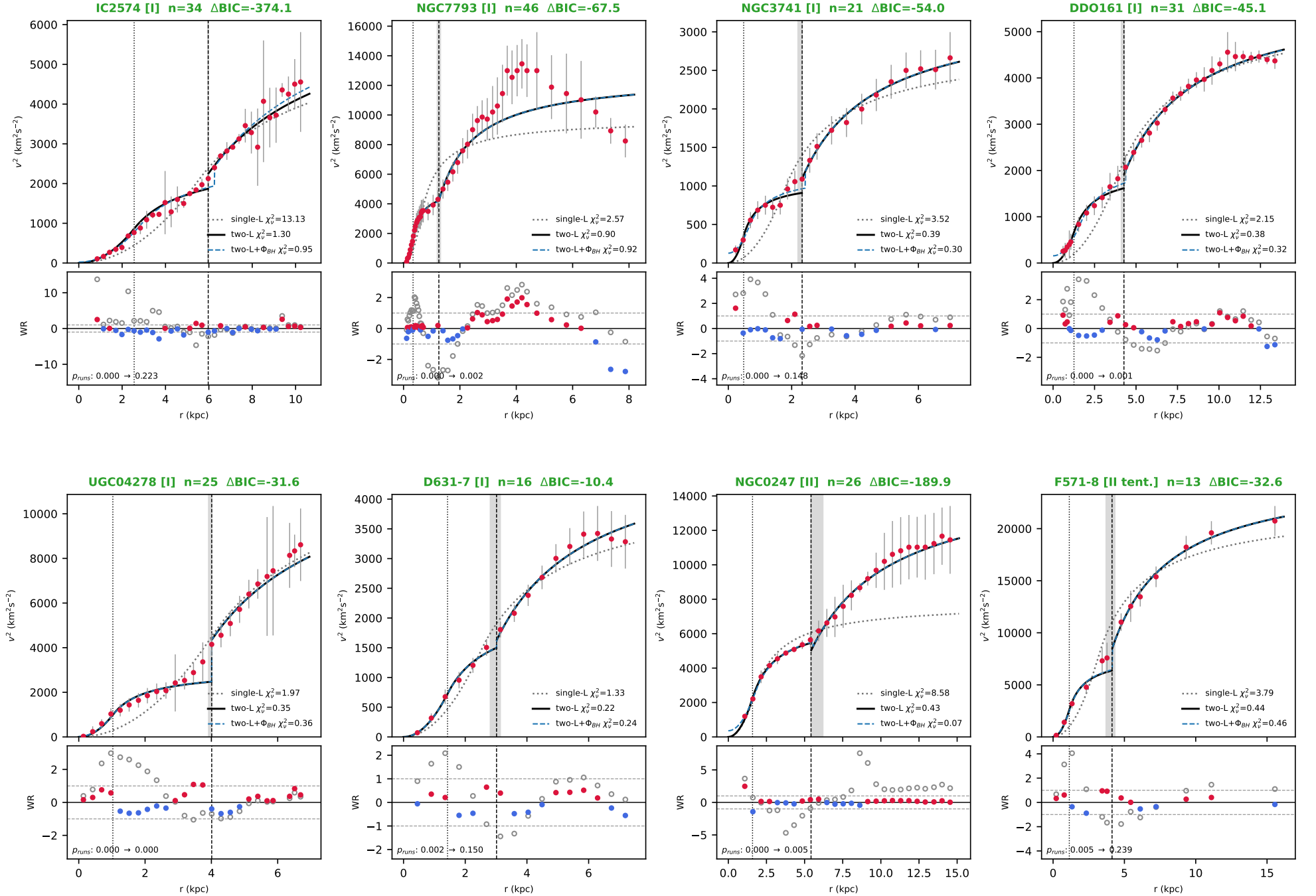
Table 5: Accepted two-L fits: goodness of fit. Subscripts 1, 2, 3 = single-L, two-L, two-L+ Φ_{BH} . $P_{<} = P(\chi^2 \leq \chi^2_{\text{obs}})$; jk = largest relative leave-one-out scatter of the four parameters.

Galaxy	χ^2_{L1}	χ^2_{L2}	χ^2_{L3}	BIC ₁	BIC ₂	BIC ₃	RMS ₂	$p_{\text{Runs},2}$	$p_{\text{Runs},3}$	DW ₂	$P_{<}$	jk
IC2574	13.13	1.30	0.95	427.1	53.0	45.2	0.107	0.223	0.348	1.41	0.873	0.17
NGC7793	2.57	0.90	0.92	120.7	53.3	56.7	0.213	0.002	0.001	0.24	0.350	0.20
NGC3741	3.52	0.39	0.30	72.9	18.8	20.0	0.134	0.148	0.148	1.37	0.012	0.01
DDO161	2.15	0.38	0.32	69.2	24.1	25.6	0.096	0.001	0.062	0.60	0.002	0.19
UGC04278	1.97	0.35	0.36	51.8	20.2	23.3	0.141	0.000	0.001	0.74	0.003	0.16
D631-7	1.33	0.22	0.24	24.2	13.8	16.6	0.049	0.150	0.150	1.33	0.003	0.21
NGC0247	8.58	0.43	0.07	212.4	22.5	17.8	0.038	0.005	0.008	1.99	0.010	0.12
F571-8	3.79	0.44	0.46	46.9	14.2	16.5	0.132	0.239	0.394	1.56	0.086	0.47
UGC12732	2.81	0.90	0.24	44.9	21.9	16.5	0.127	0.021	0.065	0.87	0.452	0.23
UGC00731	2.34	0.16	0.17	28.4	11.3	13.6	0.030	0.272	0.541	1.37	0.005	0.30
UGC07524	1.07	0.28	0.17	38.0	21.4	21.5	0.127	0.010	0.035	0.87	0.000	0.30
ESO116-G012	2.57	1.49	0.77	38.8	27.2	21.2	0.194	0.092	0.215	1.80	0.871	2.28
NGC0055	1.04	0.33	0.23	25.9	17.8	18.9	0.056	0.007	0.025	1.21	0.004	0.63
NGC5585	28.93	4.05	4.37	642.9	93.8	98.8	0.127	0.002	0.002	1.08	1.000	0.26
DDO154	7.43	3.48	3.10	79.3	37.7	34.1	0.141	0.272	0.541	0.76	0.999	0.35
NGC2403	32.04	7.93	7.10	2283.4	564.2	504.3	0.105	0.000	0.000	0.62	1.000	0.22

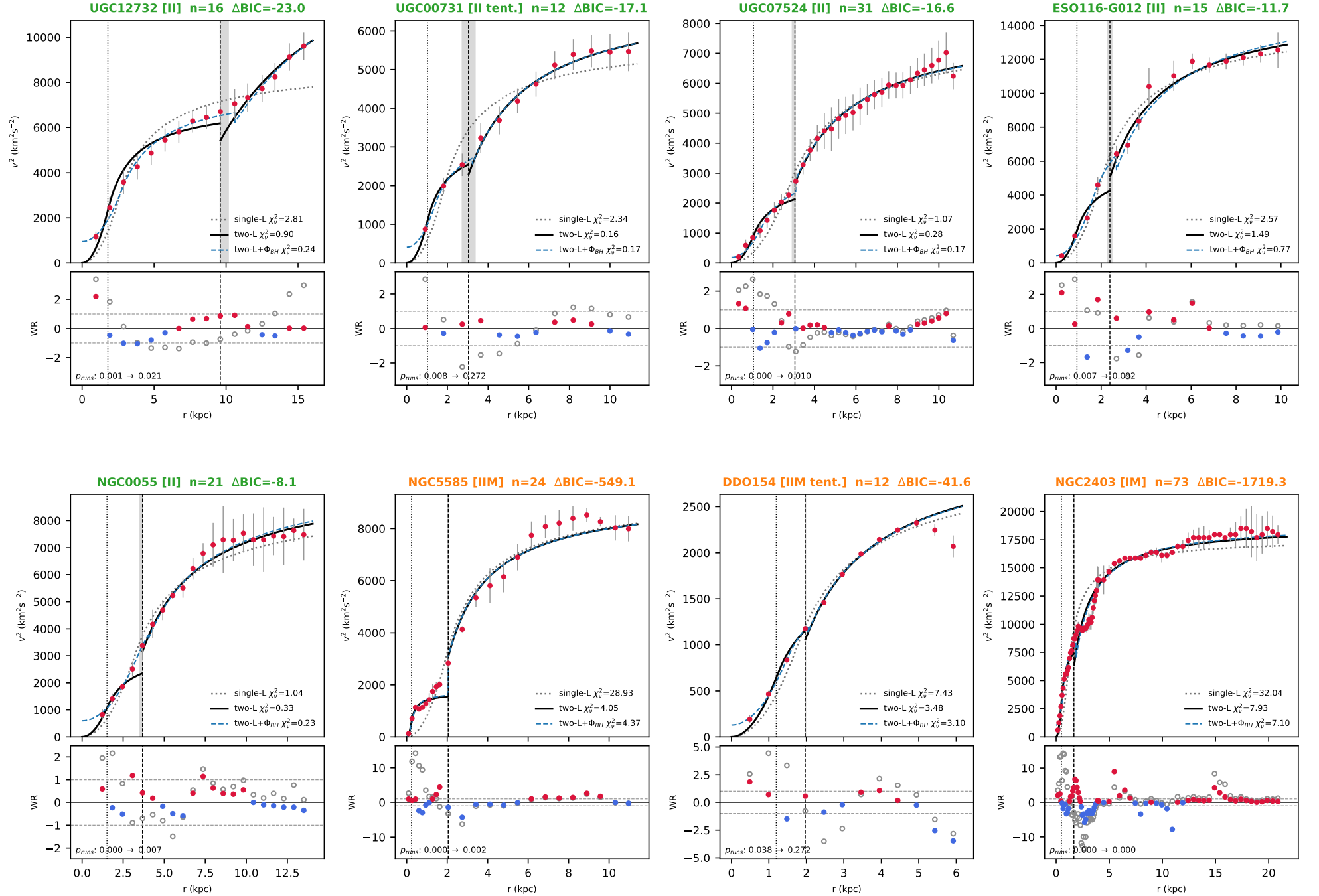
Galaxy	$\chi^2_{\nu/1}$	$\chi^2_{\nu/2}$	$\chi^2_{\nu/3}$	BIC ₁	BIC ₂	BIC ₃	RMS ₂	$p_{\text{runs},2}$	$p_{\text{runs},3}$	DW ₂	$P_{<}$	jk
UGC03580	9.69	3.29	1.51	443.6	156.7	82.8	0.337	0.000	0.000	0.40	1.000	0.19
NGC1003	4.91	2.77	1.02	174.2	103.0	49.6	0.114	0.000	0.002	0.38	1.000	0.15
UGC06787	25.98	17.26	11.01	1800.9	1173.5	748.0	0.129	0.000	0.000	0.78	1.000	0.41
NGC4013	5.52	2.36	0.48	194.8	90.0	32.7	0.118	0.000	0.000	0.48	1.000	0.08
UGC02885	2.03	0.75	1.18	40.3	23.1	31.2	0.056	0.017	0.019	1.17	0.267	0.68
UGC06614	1.62	0.33	0.22	23.0	13.2	14.6	0.054	0.007	0.197	1.20	0.033	0.43
NGC3726	2.98	2.19	1.94	34.7	27.4	26.0	0.091	0.409	0.338	2.11	0.975	0.21

Atlas 1: accepted two-L fits with residuals

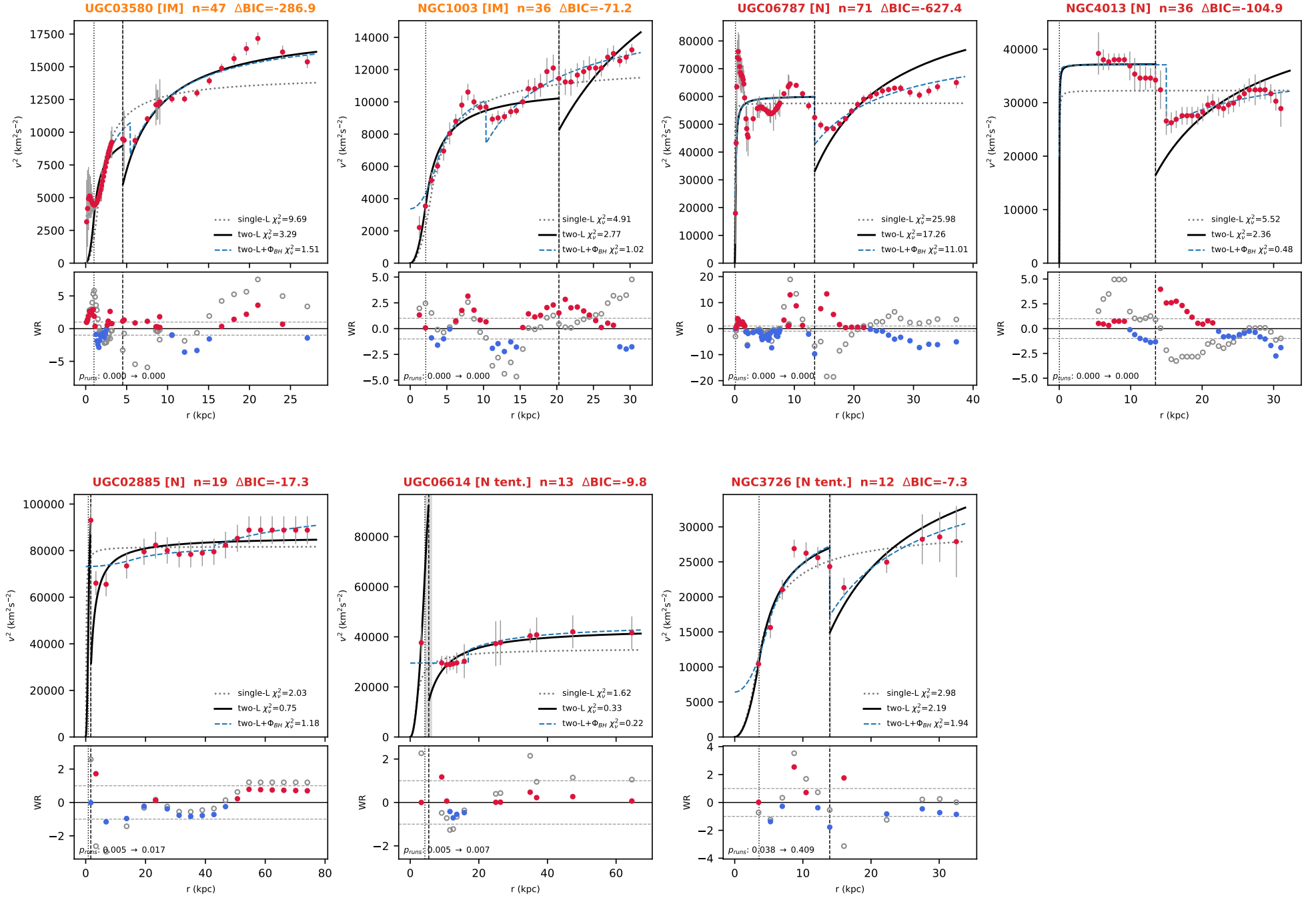
Category B, accepted two-L fits (green clean nested, orange needs more regions, red not nested). Shaded: profile $\Delta\chi^2 \leq 1$ interval of R_2 . Residuals: single-L open grey, two-L filled.



Category B, accepted two-L fits (green clean nested, orange needs more regions, red not nested). Shaded: profile $\Delta\chi^2 \leq 1$ interval of R_2 . Residuals: single-L open grey, two-L filled.

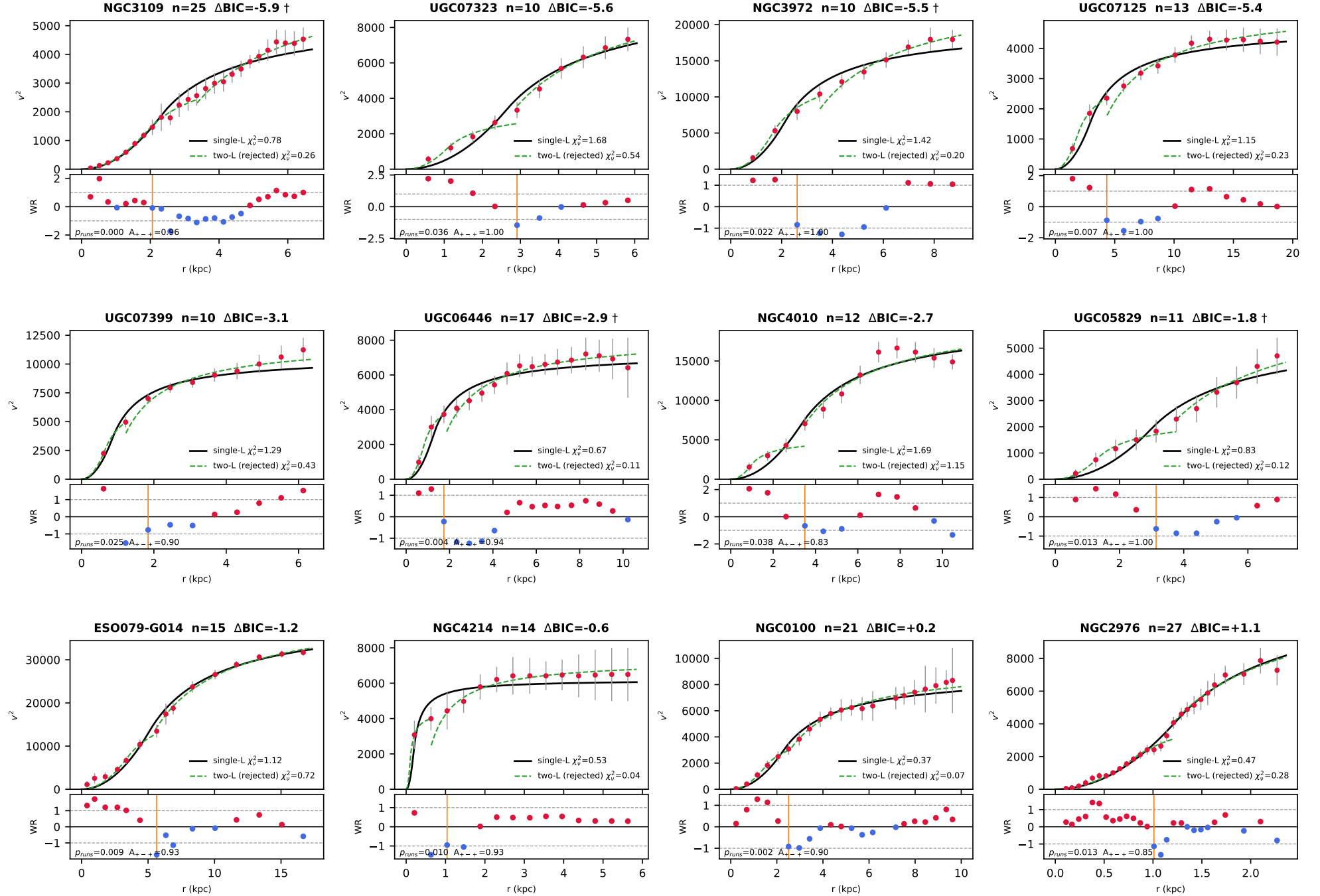


Category B, accepted two-L fits (green clean nested, orange needs more regions, red not nested). Shaded: profile $\Delta\chi^2 \leq 1$ interval of R_2 . Residuals: single-L open grey, two-L filled.



Atlas 2: category-B galaxies with single-L kept

Category B, two-L not accepted ($\Delta\text{BIC} \geq -6$): single-L fit and its residuals (orange: first crossover); dashed: best two-L for comparison. † 2018 double fit.



Category B, two-L not accepted ($\Delta\text{BIC} \geq -6$): single-L fit and its residuals (orange: first crossover); dashed: best two-L for comparison. † 2018 double fit.

